



NCA Sponsored Training

Day 1 – Morning Session (Theory)

Introduction to Cybersecurity

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Who am I?





Outline



1. Course Introduction

- *Objectives*
- *Structure*
- *Final Exam*

2. Introduction to Cybersecurity

- *What is Cybersecurity?*
- *Security Goals*
- *Common Threats*

3. Famous Real-world Attacks



Course Introduction - Objectives



The objective of this course is to expose the students to **fundamentals, applications and technologies** related to **cybersecurity**, providing the methodological basis, **skills**, and **expertise** to further progress in the cybersecurity domain.



Course Introduction - Structure



- **16** modules (8 theory, 8 hands-on) over **4** days
- **4** modules per day
- **Final exam** on the 5th day

Subjects:

- Introduction to Cybersecurity
- Cryptography
- Authentication (Password Security)
- Network Security
- Web Security
- Social Engineering



What is Security?

Security refers to **freedom** from, or **resilience** against, potential **harm** from external forces.

Beneficiaries of security may be persons and social groups, objects and institutions, ecosystems, and any other entity or phenomenon vulnerable to unwanted change by its environment.



What is Cybersecurity?

Security, in Information Technology (IT), is the **defense** of digital **information** and IT **assets** against internal and external, malicious and accidental **threats**.



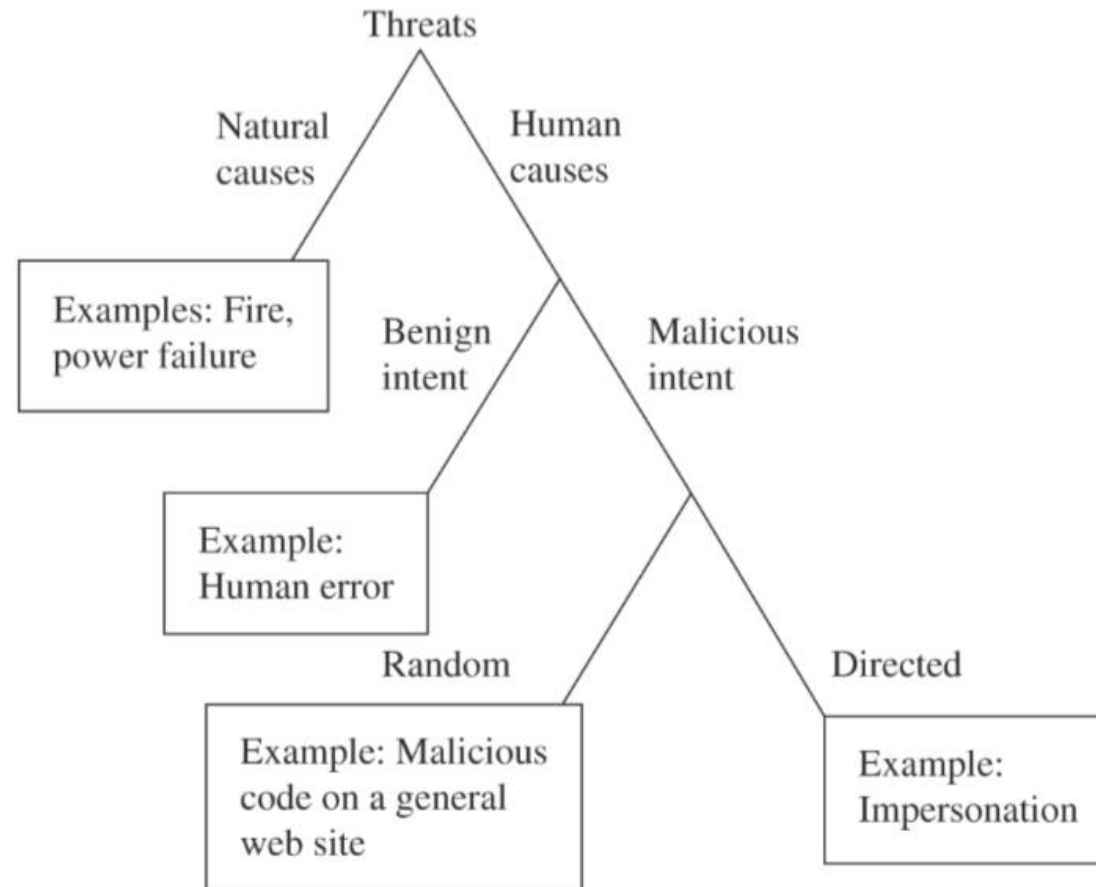
Important Terms



- **Asset:** Stuff we care about, such as information, software, hardware, bandwidth, reputation, privacy, money, etc.
- **Threat:** The potential for an occurrence that would cause an undesirable effect on an asset. Threats are often evaluated with respect to the CIA triad..
- **Vulnerability:** A weakness in a system that allows a threat to affect an asset.
- **Exploit:** A technique that takes advantage of a specific vulnerability to achieve some effect on an asset.
- **Attacker:** A human (so far) who exploits a vulnerability
- **Controls or countermeasures:** action, device, procedure, or technique to remove or reduce a vulnerability



Kinds of Threats





Why Do Computer Attacks Occur?



• Who are the attackers?

- *Criminals*
- *Crime organizations*
- *Rogue states*
- *Industrial espionage*
- *Angry employees*
- *Bored teenagers*



• Why they do it?

- *Profit*



- *Fun*



- *Fame*





Attacker Goals



Why are our systems and networks attacked?

- Steal our information or gather information
- Steal our money
- Use our hardware, software, or other assets
- Destroy or deny use of our assets (data, information systems, physical resources)
- Corrupt our information
- Harm reputations, make a statement
- Prepare for future action (e.g., botnets)
- Just to see if it can be done
- Penetration testing





Attack Phases



1. **Probe:** passive and active reconnaissance
2. **Penetrate:** gain initial access
 - *Software vulnerabilities*
 - *Weak passwords or configurations*
 - *Credential stealing, social engineering, insiders*
3. **Persist:** maintain access
 - *Compromised accounts, backdoors, rootkits, bots*
 - *Covering tracks*
4. **Propagate:** spread up and out
 - *Privilege escalation*
 - *Extend to other systems or networks*
5. **Profit:** achieve attack goals



Computer Security Issues



- **Malware**

- *Ransomware*
- *Spyware*
- *Adware*
- *Trojans*
- *Worm*
- *Rootkits*
- *Keyloggers*
- *Virus*



- **Distributed Denial of Service**

- **Social Engineering**

- *Phishing*
- *Baiting*
- *Pretexting*
- *Tailgating*
- *Honeytrap*
- *Smishing*

- **Zero-day**

- **Botnet**

- **Identity Theft**



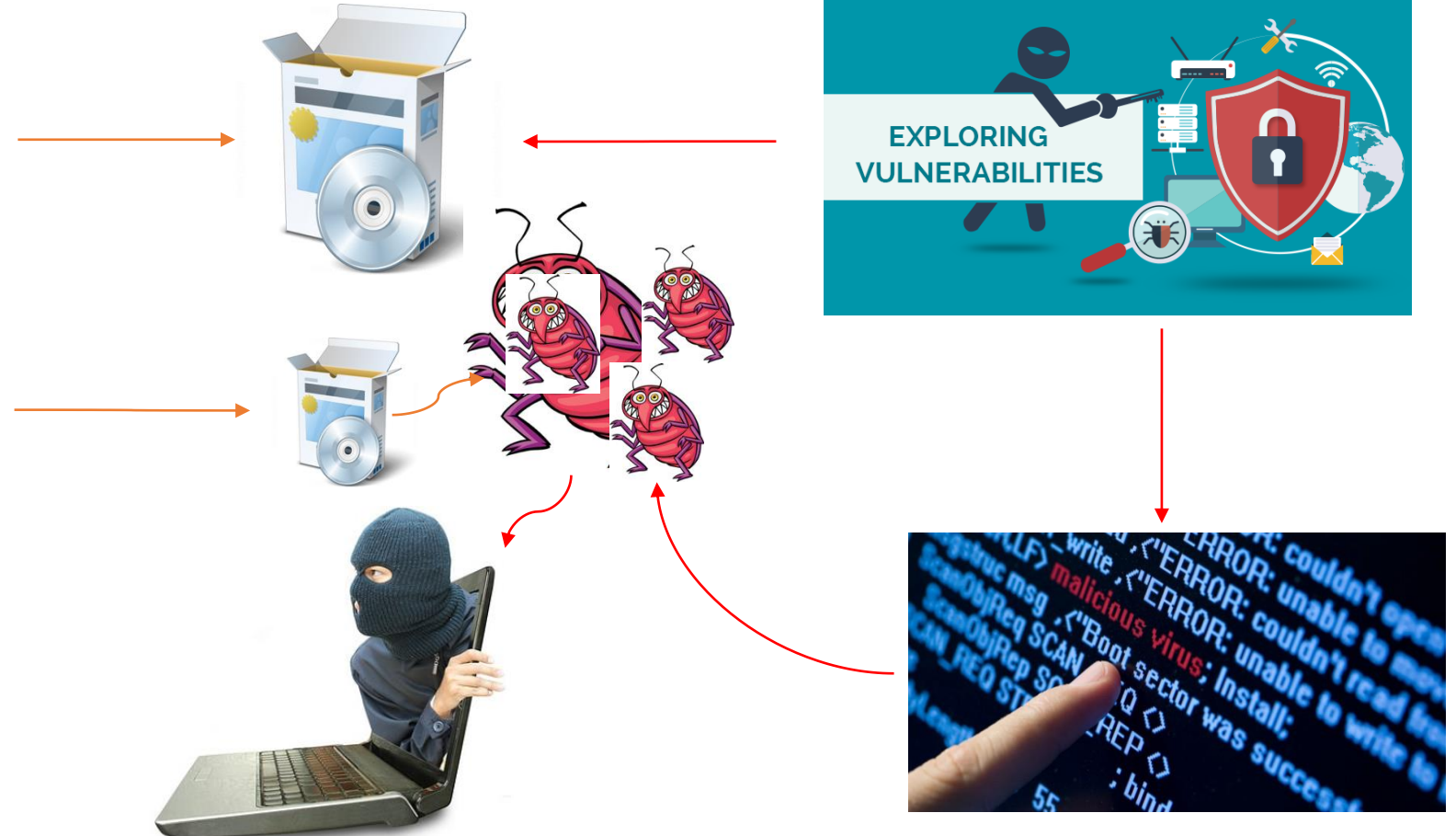
Why do these attacks happen?



- **Software/computer systems are buggy**
- **Users make mistakes**
- **Technological factors**
 - *Unsafe program languages*
 - *Software are complex, dynamic, and increasingly so*
 - *Making things secure are hard*
 - *Security may make things harder to use*



Software Developer





Why Does This Happen?



- **Economic factors**

- *Lack of incentives for secure software*
- *Security is difficult, expensive, and takes time*

- **Human factors**

- *Lack of security training for software engineers*
- *Largely uneducated population*



Cybersecurity Goals – The CIA Triad



- **Confidentiality**

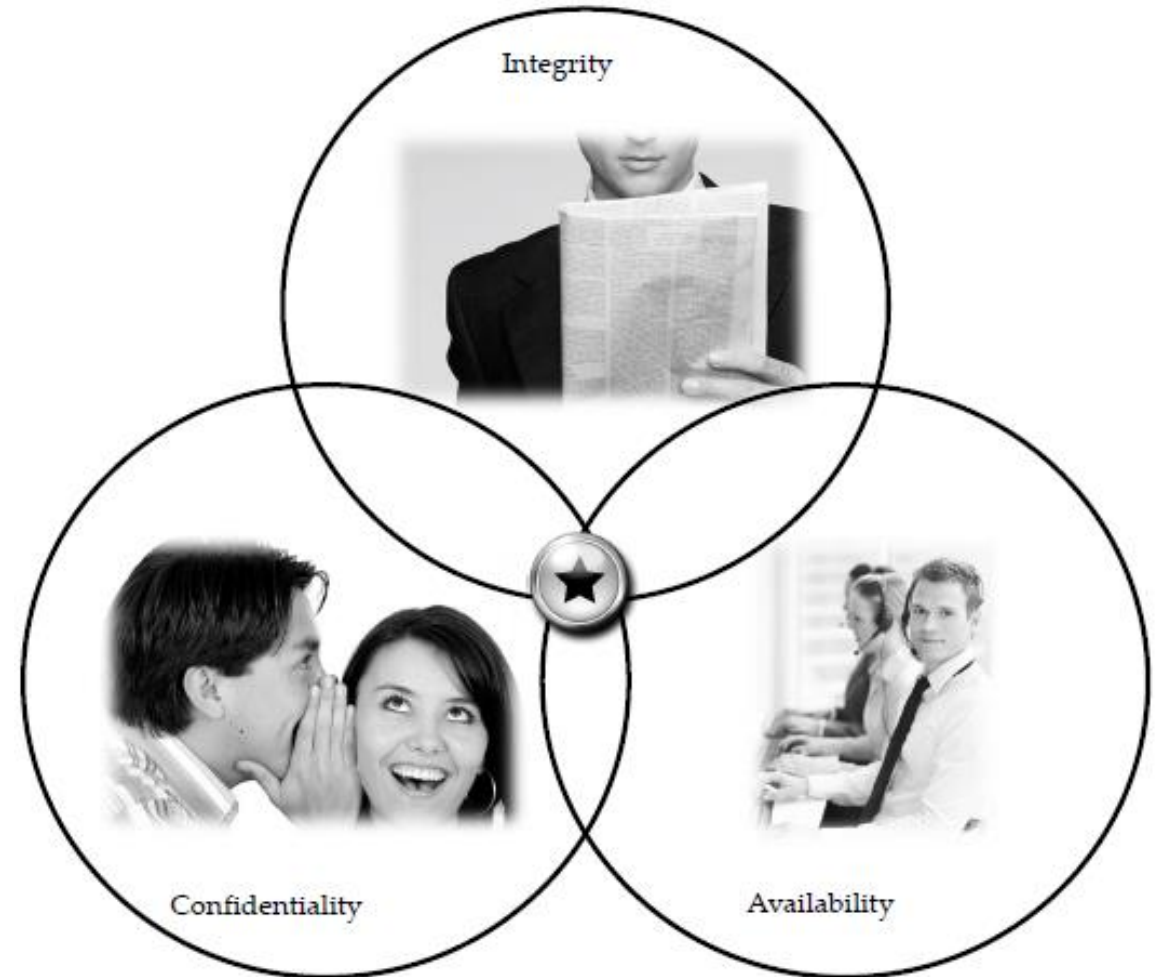
- Keeping data and resources hidden

- **Integrity**

- Data integrity (integrity)
- Origin integrity (authentication)

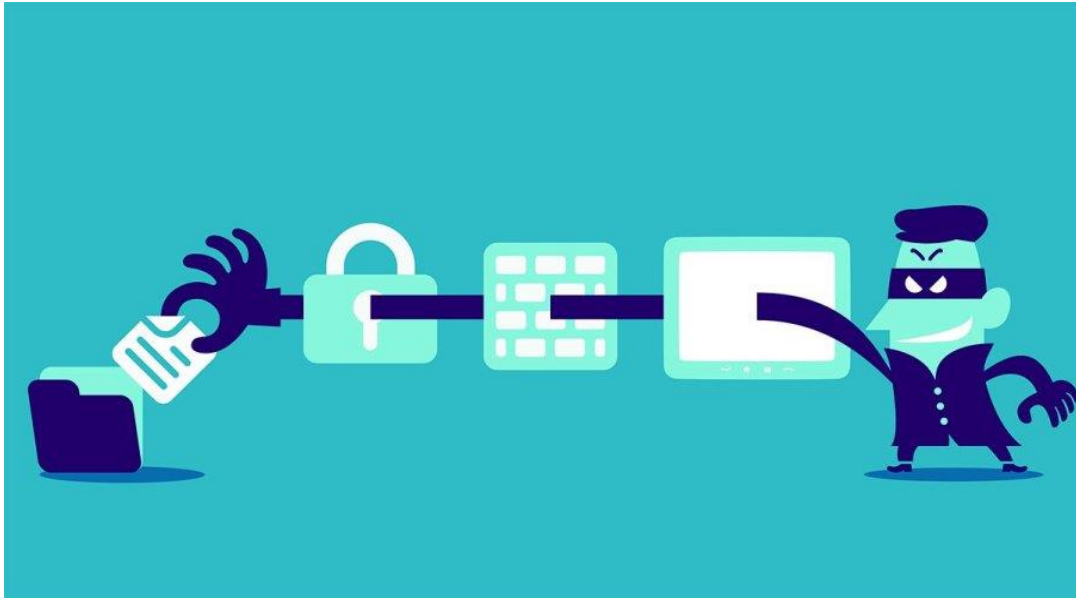
- **Availability**

- Enabling access to data and resources





Protecting **information** from disclosure to **unauthorized** entities.



How:

- Encryption
- Access Control
- Authentication



Confidentiality - Example

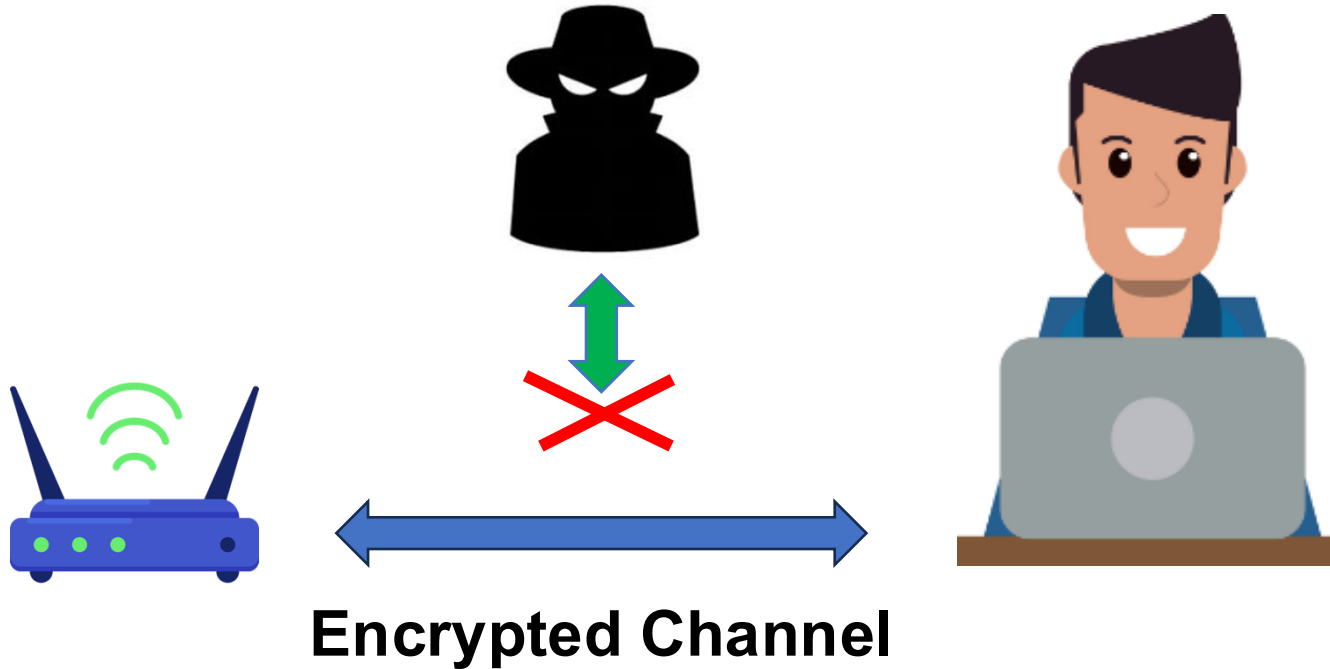


Eavesdropping





Confidentiality - Example

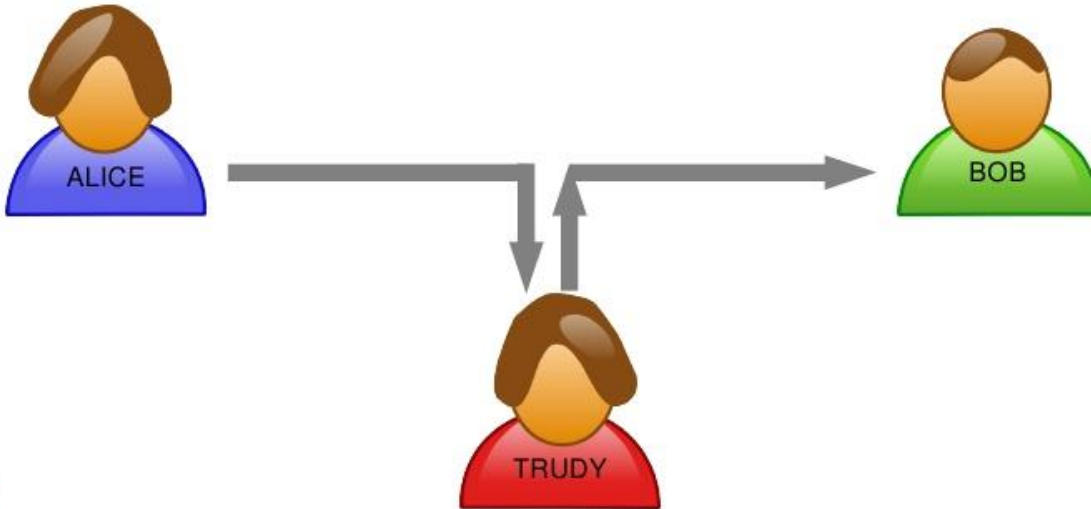


- Only use encrypted wireless channels
- WPA3 (January 2018) is the current WiFi standard
- Always use https:// (SSL/TLS) in your browser
- Use encrypted email if possible



Message Integrity

- Alice is sending a message to Bob.
- Is Bob receiving exactly what Alice is sending?



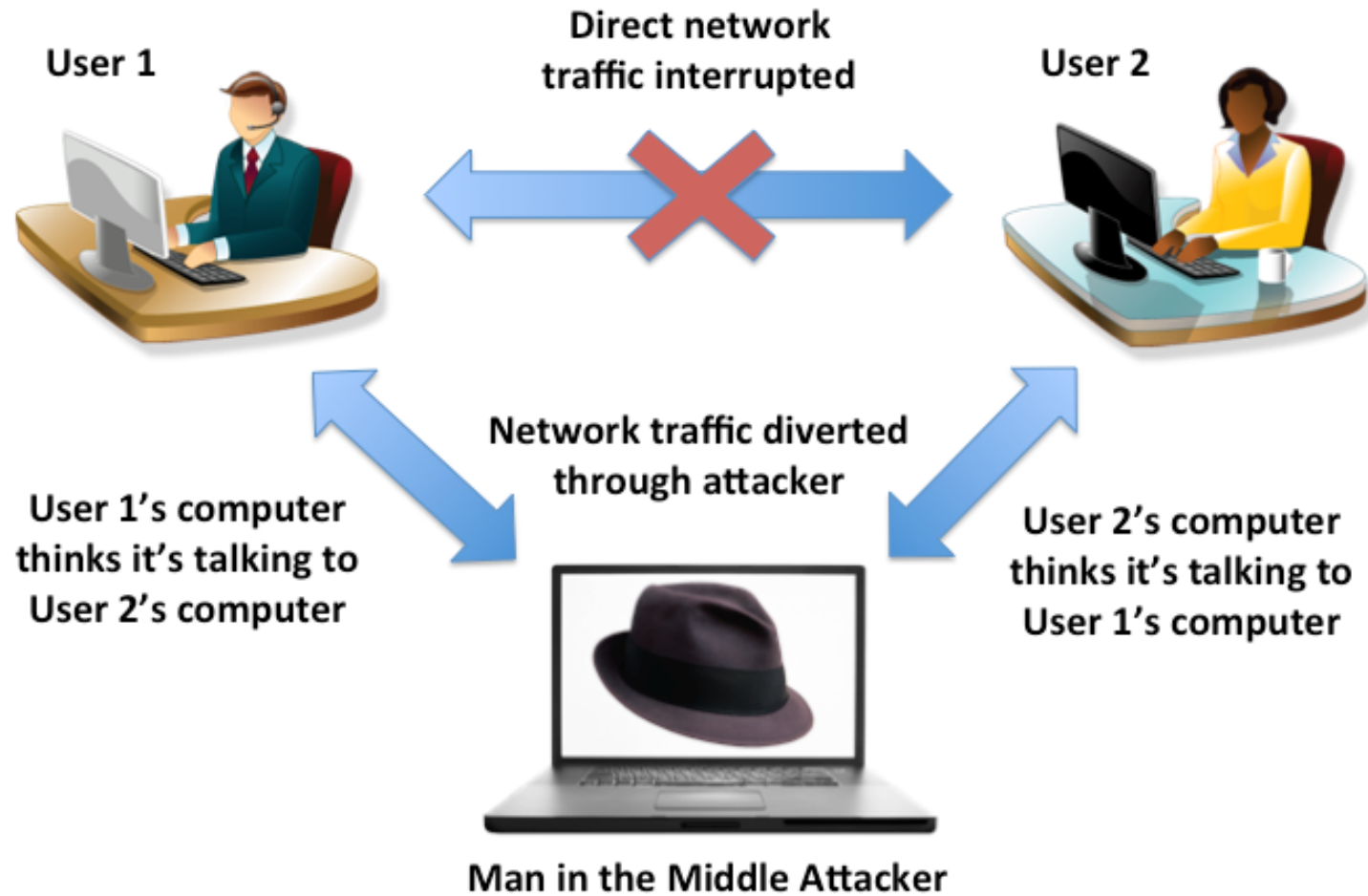
The property that **information** has not be **altered** in an unauthorized way.

How:

- Checksums
- Error Correcting Codes
- Hashing



Integrity - Example

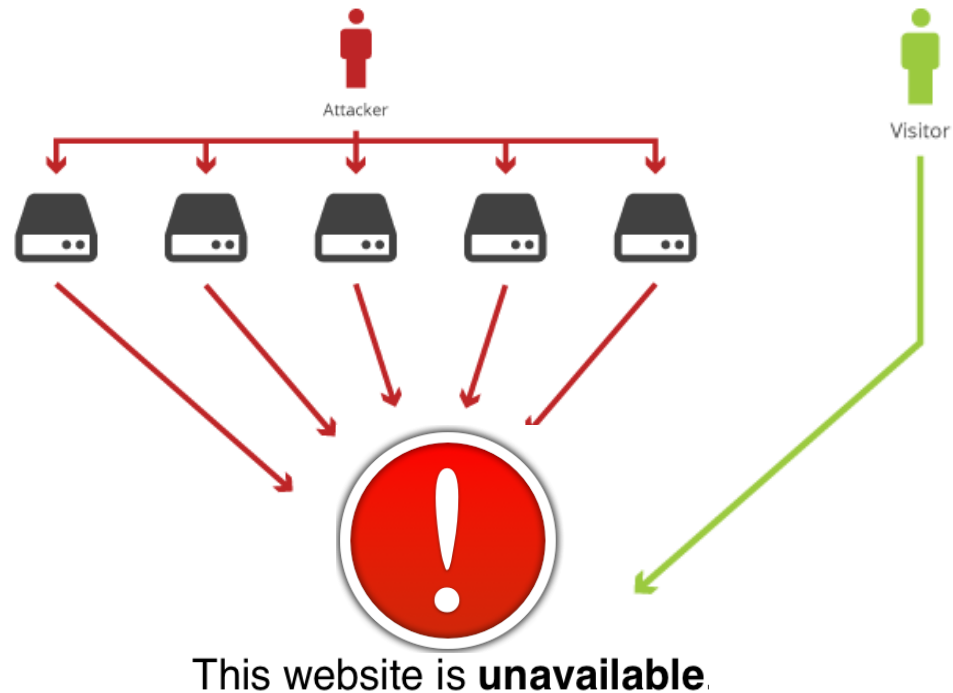




Availability



For any information system to serve its purpose, the information must be available when it is needed.



How:

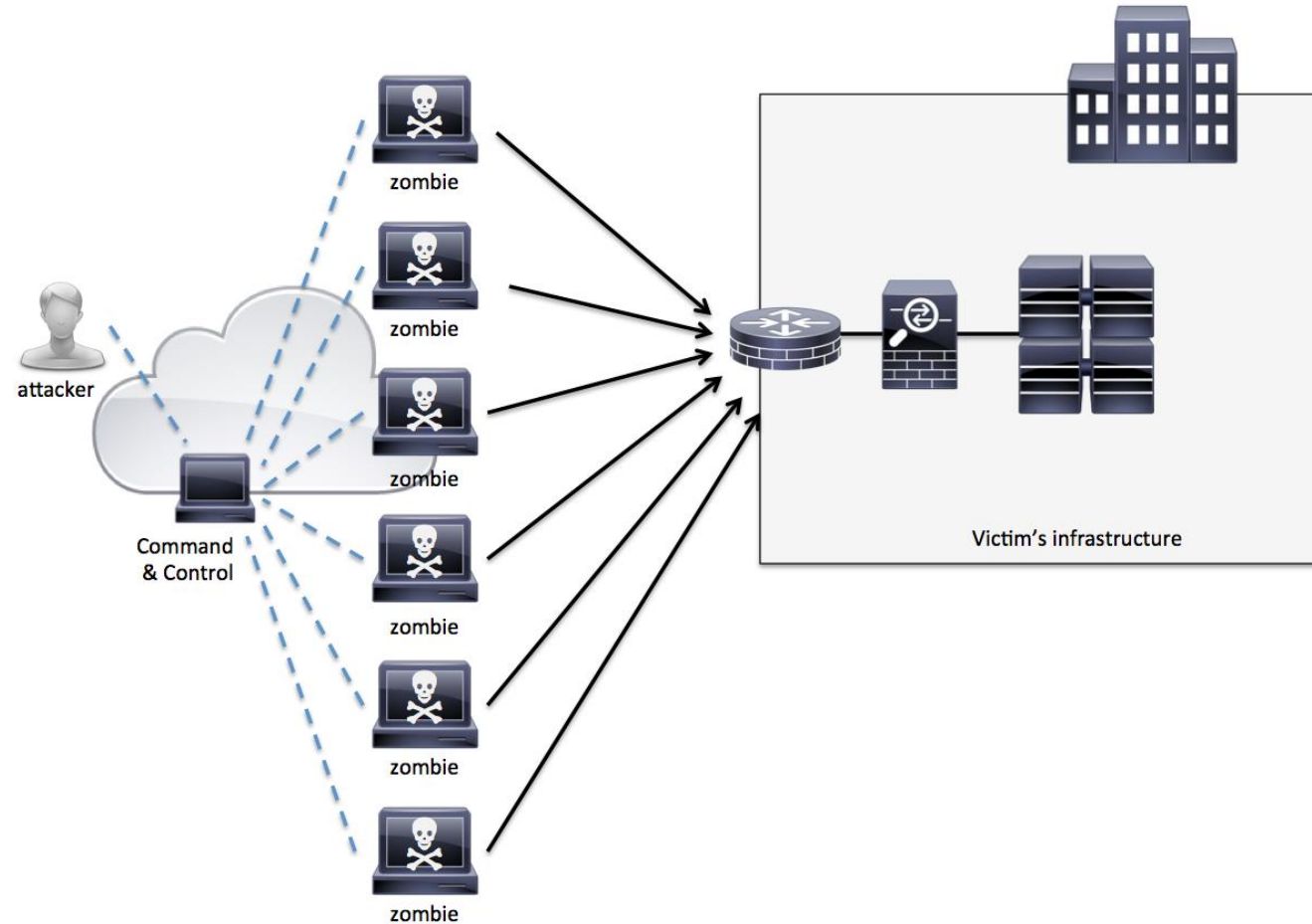
- Physical Protections
- Computational Redundancies



Availability - Example



- A cyberattack in which the perpetrator aims to make a machine or a network resource **unavailable** to its intended users.
- Leverages a network of compromised computers (BotNets, Zombie Computers) to send huge amounts of data (random data or legitimate requests) to overwhelm the target.





CIA - Recap



Example Threats	Example Countermeasures
<i>Confidentiality</i>	
Packet Sniffing	Encryption
File Grabbing	Access Control
<i>Integrity</i>	
Spoofed Email	Digital Signature
Disk Drive Corruption	Backups
<i>Availability</i>	
Denial of Service Attack	Firewall, Redundancies
Power Failure	Backup Power Supplies



More definitions



Policy

- A statement of what is and what is not allowed
- Divides the world into secure and non-secure states
- A secure system starts in a secure state. All transitions keep it in a secure state.

Mechanism

- A method, tool, or procedure for enforcing a security policy





Assurance

- Evidence of how much to trust a system
- Evidence can include
 - *System specifications*
 - *Design*
 - *Implementation*
- Mappings between the levels

Example:

- Why do you trust Aspirin from a major manufacturer?
 - *FDA certifies the aspirin recipe*
 - *Factory follows manufacturing standards*
 - *Safety seals on bottles*
- Analogy to software assurance



Type of “hackers” (most common)



Black Hat

Malicious hacker
Steal data
Monetary damage, etc



White Hat

Ethical “hacker”
Cybersecurity specialist
Protector



Red Hat

2 definitions:
One that targets Linux
Vigilante



Top Cybersecurity Threats in 2025



Increasing Attack Types

	2021	2022	2023	2024	2025
#1 Attack Type	Malware	Malware	Malware	Malware	Malware
#2 Attack Type	Ransomware	Ransomware	Ransomware	Ransomware	Phishing
#3 Attack Type	Phishing	Phishing	Phishing	Phishing	Ransomware

Top Threat Actors

	2021	2022	2023	2024	2025
#1 Threat Actor	Malicious insiders	Human error	Human error	External attackers — hacktivists	External attackers — hacktivists
#2 Threat Actor	Human error	External attackers — hacktivists	External attackers — hacktivists	Human error	External attackers — nation-state actors
#3 Threat Actor	External attackers	External attackers — nation-state actors	External attackers — nation-state actors	External attackers — nation-state actors	Human error

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Biggest Data Breaches



1) Yahoo August 2013

- 3 Billion Accounts

2) Alibaba November 2019

- 1.1 Billion pieces of user data, including usernames and mobile numbers

3) LinkedIn June 2021

- Data associated with 700 Million users

4) Weibo March 2020

- 538 Million User Accounts

5) Facebook April 2019

- Information related to more than 530 million Facebook users and included phone numbers, account names, and Facebook IDs

6) Marriot September 2018

- Data associated with 500 Million users



Biggest Ransomware Attacks



1) NotPetya 2017

- Phishing
- Estimated Monetary Impact \$10 billion

2) Wannacry 2017

- vulnerability in SMB protocol
- Estimated Monetary Impact \$4 billion

3) GrandCrab 2018/2019

- Phishing
- Estimated Monetary Impact \$2 billion

4) Locky March 2020

- phishing emails distributing a macro in a Word document
- Estimated Monetary Impact \$1 billion

5) Ryuk 2018-present

- initial compromise, usually TrickBot infection
- Estimated Monetary Impact \$150 million

6) REvil 2019/2021

- zero-day vulnerability
- Estimated Monetary Impact \$70 million



Worst Phishing Attacks in History



1) Facebook/Google Scam

- carefully crafted phishing emails with fake invoices, contracts and letters to employees at both these tech giants, falsely billing them for millions of dollars over a period of two years between 2013 to 2015.

2) NotPetya

- In June 2017, the world woke up to the most devastating cyberattack in history that spread across the planet like wildfire, ushering in a new era of cyber warfare.

3) Ukrainian Power Grid

- In December 2015, a Ukrainian electricity distribution company, became the world's first power grid provider to be taken down in a cyberattack. The threat actors were able to attack the target and force a blackout through a phishing email.

Do you have any questions?



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